

C-DAC — ASSIGNMENT 01 ANSWERS

QUESTION 01

Write a Java program to declare two integer variables and perform addition, subtraction, multiplication, division, and remainder operations. Display all the results.

LOGIC

Declare two integer variables `num1` and `num2`. Compute addition (+), subtraction (-), multiplication (*), quotient (/), and modulus (%) and display each output.

CODE

```
// Assignment 01 - Question 01: Arithmetic Operations
public class Q01 {
    public static void main(String[] args) {
        int num1 = 20;
        int num2 = 6;

        int sum = num1 + num2;
        int diff = num1 - num2;
        int product = num1 * num2;
        int quotient = num1 / num2;
        int remainder = num1 % num2;

        System.out.println("First Number: " + num1);
        System.out.println("Second Number: " + num2);
        System.out.println("Addition: " + sum);
        System.out.println("Subtraction: " + diff);
        System.out.println("Multiplication: " + product);
        System.out.println("Division: " + quotient);
        System.out.println("Remainder: " + remainder);
    }
}
```

EXPECTED OUTPUT

```
First Number: 20
Second Number: 6
Addition: 26
Subtraction: 14
Multiplication: 120
Division: 3
Remainder: 2
```

QUESTION 02

Write a Java program to store the radius of a circle in a variable and calculate its area and circumference. Formula: Area = $\pi \times r \times r$, Circumference = $2 \times \pi \times r$.

LOGIC

Store radius in `radius` (double) and pi constant in `pi`. Compute `area = pi \* radius \* radius` and `circumference = 2 \* pi \* radius`.

CODE

```
// Assignment 01 - Question 02: Area of a Circle
public class Q02 {
    public static void main(String[] args) {
        double radius = 7.0;
        double pi = 3.14159;

        double area = pi * radius * radius;
        double circumference = 2 * pi * radius;

        System.out.println("Radius of Circle: " + radius);
        System.out.println("Area of Circle: " + area);
        System.out.println("Circumference of Circle: " + circumference);
    }
}
```

EXPECTED OUTPUT

```
Radius of Circle: 7.0
Area of Circle: 153.93791
Circumference of Circle: 43.98226
```

QUESTION 03

Write a Java program to accept principal amount, rate of interest and time using variables and calculate the simple interest and total amount. Formula: Simple Interest = $(P \times R \times T) / 100$, Amount = P + Simple Interest.

LOGIC

Declare variables for `principal`, `rate`, and `time`. Compute `simpleInterest = (P \* R \* T) / 100` and `totalAmount = principal + simpleInterest`.

CODE

```
// Assignment 01 - Question 03: Simple Interest
public class Q03 {
    public static void main(String[] args) {
        double principal = 10000.0;
        double rate = 5.5;
        double time = 3.0; // in years

        double simpleInterest = (principal * rate * time) / 100;
        double totalAmount = principal + simpleInterest;

        System.out.println("Principal Amount: Rs. " + principal);
        System.out.println("Rate of Interest: " + rate + "%");
        System.out.println("Time Period: " + time + " years");
        System.out.println("Simple Interest: Rs. " + simpleInterest);
        System.out.println("Total Amount: Rs. " + totalAmount);
    }
}
```

EXPECTED OUTPUT

```
Principal Amount: Rs. 10000.0
Rate of Interest: 5.5%
Time Period: 3.0 years
Simple Interest: Rs. 1650.0
Total Amount: Rs. 11650.0
```

QUESTION 04

Write a Java program to convert temperature from Celsius to Fahrenheit. Formula: Fahrenheit = (Celsius $\times$ 9 / 5) + 32.

LOGIC

Store Celsius value in variable `celsius`. Convert to Fahrenheit using `(celsius \* 9.0 / 5.0) + 32.0`.

CODE

```
// Assignment 01 - Question 04: Temperature Conversion
public class Q04 {
    public static void main(String[] args) {
        double celsius = 37.0;

        double fahrenheit = (celsius * 9.0 / 5.0) + 32.0;

        System.out.println("Temperature in Celsius: " + celsius + " deg C");
        System.out.println("Temperature in Fahrenheit: " + fahrenheit + " deg F");
    }
}
```

EXPECTED OUTPUT

```
Temperature in Celsius: 37.0 deg C
Temperature in Fahrenheit: 98.6 deg F
```

QUESTION 05

Write a Java program to store marks of three subjects in variables and calculate the total and average marks.

LOGIC

Store marks of three subjects in `subject1`, `subject2`, `subject3`. Calculate `total = subject1 + subject2 + subject3` and `average = total / 3.0`.

CODE

```
// Assignment 01 - Question 05: Calculate Total and Average
public class Q05 {
    public static void main(String[] args) {
        int subject1 = 85;
        int subject2 = 90;
        int subject3 = 78;

        int total = subject1 + subject2 + subject3;
        double average = total / 3.0;

        System.out.println("Subject 1 Marks: " + subject1);
        System.out.println("Subject 2 Marks: " + subject2);
        System.out.println("Subject 3 Marks: " + subject3);
        System.out.println("Total Marks: " + total);
        System.out.println("Average Marks: " + average);
    }
}
```

EXPECTED OUTPUT

```
Subject 1 Marks: 85
Subject 2 Marks: 90
Subject 3 Marks: 78
Total Marks: 253
Average Marks: 84.33333333333333
```

QUESTION 06

Write a Java program to accept an integer and check whether the number is even or odd using if-else.

LOGIC

Check `number % 2 == 0`. If true, the number is Even; otherwise, it is Odd.

CODE

```
// Assignment 01 - Question 06: Even or Odd
public class Q06 {
    public static void main(String[] args) {
        int number = 15;

        System.out.println("Given Number: " + number);
        if (number % 2 == 0) {
            System.out.println(number + " is an Even number.");
        } else {
            System.out.println(number + " is an Odd number.");
        }
    }
}
```

EXPECTED OUTPUT

```
Given Number: 15
15 is an Odd number.
```

QUESTION 07

Write a Java program to accept an integer and check whether it is positive, negative, or zero using if and if-else.

LOGIC

Use conditional branching: check `number > 0` (Positive), `number < 0` (Negative), else Zero.

CODE

```
// Assignment 01 - Question 07: Positive, Negative or Zero
public class Q07 {
    public static void main(String[] args) {
        int number = -12;

        System.out.println("Given Number: " + number);
        if (number > 0) {
            System.out.println(number + " is Positive.");
        } else if (number < 0) {
            System.out.println(number + " is Negative.");
        } else {
            System.out.println("The number is Zero.");
        }
    }
}
```

EXPECTED OUTPUT

```
Given Number: -12
-12 is Negative.
```

QUESTION 08

Write a Java program to accept two numbers and find the larger number using if-else. If both numbers are equal, display an appropriate message.

LOGIC

Compare `num1` and `num2` using `if-else-if`. Display larger number or equality message.

CODE

```
// Assignment 01 - Question 08: Largest of Two Numbers
public class Q08 {
    public static void main(String[] args) {
        int num1 = 45;
        int num2 = 78;

        System.out.println("Number 1: " + num1);
        System.out.println("Number 2: " + num2);

        if (num1 > num2) {
            System.out.println(num1 + " is larger than " + num2);
        } else if (num2 > num1) {
            System.out.println(num2 + " is larger than " + num1);
        } else {
            System.out.println("Both numbers are equal.");
        }
    }
}
```

EXPECTED OUTPUT

```
Number 1: 45
Number 2: 78
78 is larger than 45
```

QUESTION 09

Write a Java program to accept a person's age and check whether the person is eligible to vote (Age $\geq$ 18).

LOGIC

Check condition `age  $\geq$  18`. Display 'Eligible to vote' if true, else 'Not eligible'.

CODE

```
// Assignment 01 - Question 09: Voting Eligibility
public class Q09 {
    public static void main(String[] args) {
        int age = 20;

        System.out.println("Age: " + age);
        if (age  $\geq$  18) {
            System.out.println("Eligible to vote.");
        } else {
            System.out.println("Not eligible to vote.");
        }
    }
}
```

EXPECTED OUTPUT

```
Age: 20
Eligible to vote.
```

QUESTION 10

Write a Java program to accept marks of a student and check whether the student has passed or failed (Marks >= 40).

LOGIC

Check condition `marks >= 40`. Display 'Pass' if true, else 'Fail'.

CODE

```
// Assignment 01 - Question 10: Student Result
public class Q10 {
    public static void main(String[] args) {
        int marks = 65;

        System.out.println("Student Marks: " + marks);
        if (marks >= 40) {
            System.out.println("Result: Pass");
        } else {
            System.out.println("Result: Fail");
        }
    }
}
```

EXPECTED OUTPUT

```
Student Marks: 65
Result: Pass
```
